A Relational Database for Brazilian Jiu-Jitsu Match Tracking

Student number: 3520872

***Abstract*—Write your abstract here.**

I. Introduction

Brazilian Jiu-Jitsu(BJJ) is a grappling martial art which consists of 1 vs 1 matches between fighters, in typical competitions fighters can win their matches via submissions, points, and various other methods. This means tracking fighter's gyms and results in a single spreadsheet can quickly lead to repeated data and inconsistent records.Therefore, the aim of this project is to tackle this problem by managing scenarios as a relational database that can answer questions such as who won a match,their method of victory, and how often each fighter wins.

The database consists of five entities: gym, fighter, VicotryType, match, FighterMatchPerformance.This is because a fighter takes part in many matches which involve two fighters, FighterMatchPerfomance resolves this many-to-many relationship by using a composite primary key that is made of both FighterID and MatchID.Win rate is a derived value that is calculated by a query rather than being stored, this avoids redundancy and out of date values.The database design assumes that each match has two performance records,meaning that each fighter belongs to one gym, each match has one victory method, and all data are fictional.Section II will present the design, Section III the implementation, Section IV the queries, and Section V will discuss the complexity.